

Muneeb Musharaf

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EDUCATION

University College London

MSc Robotics and Artificial Intelligence

London, UK

Starting October 2026

- Selected for the **Martingale Scholarship**, a competitive, interview-based award funding the MSc in AI and Robotics at UCL.

Queen Mary University of London

BSc Computer Science and Mathematics

London, UK

Sept 2023 – July 2026

- First Class Honours**
- Related Modules:** Procedural & Object-oriented Programming, Automata & Formal Languages, Linear Algebra, Data structures & Algorithms, Software Engineering, Internet Protocols, Probability & Statistics, Analysis of Calculus, Numbers & Functions, Vectors & Matrices, Complex Variables, Big Data Processing, Differential Equations, Complexity & Algorithms, Bayesian Decision & Risk Analysis, Neural Networks, Complex Networks

EXPERIENCE

Google DeepMind

Research Placement

London, UK

June 2025 – Aug 2025

- Selected as **1 of 12** candidates for a competitive AI research program by **Google DeepMind** with PhD supervision.
- Researched **AI perception of music** using timbre-based tasks by implementing **zero-shot and MLP experiments**.
- Resulted in **co-authoring a research paper**, which was accepted into **the ISMIR 2025** conference [here](#).

Data Annotation

Freelance Programmer and Annotator

London, UK

Feb 2025 – Present

- Provide expert solutions to mathematical problems and provide feedback to **train AI models** and complex data sets.
- Program projects in a variety of coding languages depending on the specifications.

Private Tutoring Practice

Physics and Mathematics Tutor

London, UK

Aug 2023 – Present

- Delivered personalized tutoring and designed worksheets in Mathematics and Physics from GCSE to A-level.
- Ensured **90%** of students achieved high grades through tailored one-to-one and group sessions.

Queen Mary Ihsan Consulting Society

Committee member, Marketing Manager, Tech Consulting Co-Lead

London, UK

Oct 2024 - Present

- Consult small businesses and charities on marketing strategy, managing social media campaigns through **Canva** that increased applications by **20%**.
- Co-lead the society's **tech consulting** by managing small teams to deliver **software** and **AI solutions** for clients.
- Led a six-person team to build an **AI-powered ethical screening tool** for **Kestrl** (an Islamic banking app) evaluating equities against Sharia-compliant investment equities.

Microsoft Accelerator

Internship

London, UK

Oct 2024 – Dec 2024

- Participated in Microsoft's structured mentoring programme, gaining direct exposure to industry practices in **software development** and **technology consulting**.
- Applied technical and collaborative skills to real-world problem-solving scenarios under professional guidance.

PROJECTS

Virtual Mars Gaussian Splatting

Final Year project

Aug 2025 – May 2026

- Investigated **3D Gaussian Splatting (3DGS)** for reconstructing Martian terrain from NASA Curiosity rover imagery, benchmarking a **COLMAP Structure-from-Motion pipeline** against a **CAHVOR-direct camera pose pipeline**.
- Diagnosed a camera intrinsic calibration error as the dominant factor limiting reconstruction quality, correcting it to raise PSNR by **~15 dB to 23.71 dB**.
- Validated pipeline generalisation, producing coherent 3D reconstructions to support accessible, interactive exploration of Mars surface imagery.

Cryptocurrency Trading Platform

University Group project

Jan 2025 – Aug 2025

- Led development of a full stack platform using **Django** and **React**, implementing two-factor authentication, trading circuit breakers, portfolio tracking, and real-time API price feeds.

- Followed **Agile workflows** with **UML documentation**, focused on database integration for balances, deposits, withdrawal, and mock trade execution.

Tetris Project

Personal project

Jun 2024 – Aug 2024

- Developed a Tetris game in **Java** using the **Javax** library to deepen my understanding of Java and interface design.
- Incorporated sounds and visual effects to enhance user engagement and integrated version control with Git.

ADDITIONAL SKILLS

- **Languages:** Java, Python, JavaScript, TypeScript, SQL, HTML/CSS, R, MATLAB
- **Frameworks & Libraries:** Django, Flask, FastAPI, React
- **AI & Data:** PyTorch, TensorFlow, NumPy, Pandas, Scikit-learn, Matplotlib, Neural Networks, Zero-shot Learning, Jupyter Notebooks, Google Colab
- **Tools & Platforms:** Git, GitHub, Docker, VS Code, IntelliJ, PyCharm, Visual Paradigm, PostgreSQL, MongoDB, Redis, RESTful APIs, WebSockets, Canva, Google Ads, Facebook Ads, Microsoft Office (Excel, Word, PowerPoint, Teams)
- **Operating Systems:** Windows, MacOS